Cerebral Hemodynamics Underlying Artery-to-Artery Embolism in Symptomatic Intracranial Atherosclerotic Disease

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Abstract

Artery-to-artery embolism (AAE) is a common stroke mechanism in intracranial atherosclerotic disease (ICAD), associated with a considerable risk of recurrent stroke. We aimed to investigate cerebral hemodynamic features associated with AAE in symptomatic ICAD. Patients with anterior-circulation, symptomatic ICAD confirmed in CT angiography (CTA) were recruited. We classified probable stroke mechanisms as isolated parent artery atherosclerosis occluding penetrating artery, AAE, hypoperfusion, and mixed mechanisms, largely based on infarct topography. CTA-based computational fluid dynamics (CFD) models were built to simulate blood flow across culprit ICAD lesions. Translesional pressure ratio (PR = Pressure /Pressure) and wall shear stress ratio (WSSR = WSS /WSS) were calculated, to reflect the relative, translesional changes of the two hemodynamic metrics. Low PR (PR ≤ median) and high WSSR (WSSR ≥ 4th quartile) respectively indicated large translesional pressure and elevated WSS upon the lesion. Among 99 symptomatic ICAD patients, 44 had AAE as a probable stroke mechanism, 13 with AAE alone and 31 with coexisting hypoperfusion. High WSSR was independently associated with AAE (adjusted OR = 3.90; P = 0.022) in multivariate logistic regression. There was significant WSSR-PR interaction on the presence of AAE (P for interaction = 0.013): high WSSR was more likely to associate with AAE in those with low PR (P = 0.075), but not in those with normal PR (P = 0.959). Excessively elevated WSS in ICAD might increase the risk of AAE. Such association was more prominent in those with large translesional pressure gradient. Hypoperfusion, commonly coexisting with AAE, might be a therapeutic indicator for secondary stroke prevention in symptomatic ICAD with AAE. [Abstract copyright: © 2023. The Author(s), under exclusive licence to Springer Science+Business Media, LLC, part of Springer Nature.]

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Keywords

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- Hemodynamics

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